Arista Access Node KR-ACC-20 Routing Protocol Flap Troubleshooting and Resolution SOP

Node: KR-ACC-20 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8877

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve the Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on the Arista Access Node KR-ACC-20 at KR Puram Access. The failure is characterized by the alarm code ALARM_8877 with a severity of Critical. The underlying cause of this failure is the misconfiguration of the routing protocols or hardware issues such as faulty network interfaces, which lead to a high rate of route flaps, causing OSPF adjacency loss, ISIS LSP flooding, and BGP route churn. The Access Node KR-ACC-20 is a critical component of the network infrastructure, and its failure can have a significant impact on network performance and availability. The SOP is designed to help network engineers quickly identify the root cause of the failure and resolve it in a timely manner to minimize downtime and ensure network reliability.

Pre-requisite Checks

- Check: Verify that the redundant link status is operational and functioning correctly.

- Check: Ensure that the remote hands team is available to dispatch to KR Puram Access for any physical interventions required.

- Check: Verify that the network management system (NMS) and other monitoring tools are functioning correctly and providing accurate alarms and notifications.

Potential Root Causes

Cause	Probability
Misconfigured routing protocols	High
Faulty network interfaces	Medium
Insufficient network resources (CPU, memory, buffers)	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 BGP router identifier 10.10.10.10, local AS number 64512 BGP table version is 1, main routing table version is 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State Pri, Description: This command displays a summary of the BGP routing table and configuration. It will help identify any issues with BGP neighbors, routing table version, and main routing table version.

Command: show ospf neighbor, Expected Output: OSPF Process 100 ID 10.10.10.10 Router ID 10.10.10.1 (10.10.10.1) Neighbor ID 10.10.10.2
```

```
(10.10.10.2) State 2WAY, State 2WAY Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec , Description: This command displays information about the OSPF neighbors, their states, and the timers associated with them. It will help identify any issues with OSPF adjacency.

Command: show isis database, Expected Output: ISIS Routing Database (Level 2) Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 , Description: This command displays the ISIS routing database, including the level 2 metric. It will help identify any issues with ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify and correct the routing protocol configuration, Command: show running-config | include bgp|ospf|isis, Description: Verify that the routing protocol configuration is correct and make any necessary corrections to prevent route flaps.

Step 2: Step: Check and replace faulty network interfaces, Command: show interface, Description: Check the status of all network interfaces and replace any faulty interfaces to prevent further issues.

Step 3: Step: Check and adjust network resources, Command: show system resources, Description: Check the system resources (CPU, memory, buffers) and adjust as necessary to prevent resource exhaustion and route flaps.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 BGP router identifier 10.10.10.10, local AS number 64512 BGP table version is 1, main routing table version is 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State Pri, Description: Verify that the BGP routing table and configuration are correct.

Command: show ospf neighbor, Expected Output: OSPF Process 100 ID 10.10.10.10 Router ID 10.10.10.1 (10.10.10.1) Neighbor ID 10.10.10.2 (10.10.10.2) State 2WAY, State 2WAY Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec Timer 5.3 sec , Description: Verify that the OSPF neighbors and their states are correct.

Command: show isis database, Expected Output: ISIS Routing Database (Level 2) Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 Metric: 1000 , Description: Verify that the ISIS routing database is correct and that there are no LSP flooding issues.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy running-config startup-config, Description: Restore the original configuration to prevent further issues.

- Step: Verify the system resources, Command: show system resources, Description: Verify that the system resources are sufficient to prevent resource exhaustion and route flaps.

- Step: Verify the network interfaces, Command: show interface, Description: Verify that all network interfaces are operational and functioning correctly.